# scpPower Supporting Information

Youshan Heng

## Contents


- S1. Fidelity
- S2. Validation
- S3. Diagnostic J
- S4. Power grid
- S5. Worked examples
- S6. Pilot protocol
- S7. Registered record


The full CSV tables, original plots, manifests and registered record are deposited in the accompanying SI directory. This PDF contains one overview table per section; SHA-256 checksums and source paths are in SI/manifest.json.

## S1

S1, fidelity tables for the five series under the final simulator and the earlier failed version.

| Series | Final | Checks | Earlier | Earlier_checks |
|---|---|---|---|---|
| Specht TMT | PASS | 16/16 | FAIL | 3/16 |
| Leduc TMT | PASS | 16/16 | FAIL | 9/16 |
| Derks timsTOF | PASS | 16/16 | FAIL | 12/16 |
| Derks Q Exactive | PASS | 16/16 | FAIL | 14/16 |
| Brunner label-free | PASS | 11/11 | PASS | 11/11 |

# S2

S2, semi-synthetic validation tables and plots, first attempt and rerun.

| Series | First | First_max_gap_pp | Rerun | Rerun_max_gap_pp |
|---|---|---|---|---|
| Specht TMT | 67/72 | 19.51 | 72/72 | 3.32 |
| Leduc TMT | 66/69 | 12.81 | 69/69 | 2.35 |
| Derks timsTOF | Not evaluable | — | Not rerun | — |
| Derks Q Exactive | 6/6 | 0.39 | Not rerun | — |
| Brunner label-free | 57/57 | 5.40 | Not rerun | — |

## S3

S3, the TMT mask diagnostic.

| Series | PSM_finite | Endpoint_finite | Gap_pp | Amendment_I |
|---|---|---|---|---|
| Specht TMT | 0.3019 | 0.2041 | 9.78 | TRUE |
| Leduc TMT | 0.3484 | 0.2776 | 7.08 | TRUE |

# S4

S4, the full registered power grid with false discovery rate, rejection counts and instrument hours.

| Series | Fold | k | Max_power | n | m | Hours |
|---|---|---|---|---|---|---|
| Specht TMT | 1.25 | 1.00 | 0.0133 | 12 | 64 | 202.7 |
| Specht TMT | 1.25 | 0.75 | 0.0009 | 12 | 32 | 101.3 |
| Specht TMT | 1.25 | 0.50 | 0.0001 | 12 | 64 | 202.7 |
| Specht TMT | 1.50 | 1.00 | 0.2028 | 12 | 64 | 202.7 |
| Specht TMT | 1.50 | 0.75 | 0.0968 | 12 | 64 | 202.7 |
| Specht TMT | 1.50 | 0.50 | 0.0046 | 12 | 64 | 202.7 |
| Specht TMT | 2.00 | 1.00 | 0.3446 | 12 | 64 | 202.7 |
| Specht TMT | 2.00 | 0.75 | 0.2903 | 12 | 64 | 202.7 |
| Specht TMT | 2.00 | 0.50 | 0.1413 | 12 | 64 | 202.7 |
| Leduc TMT | 1.25 | 1.00 | 0.0396 | 12 | 64 | 174.2 |
| Leduc TMT | 1.25 | 0.75 | 0.0021 | 12 | 64 | 174.2 |
| Leduc TMT | 1.25 | 0.50 | 0.0001 | 12 | 64 | 174.2 |
| Leduc TMT | 1.50 | 1.00 | 0.3081 | 12 | 64 | 174.2 |
| Leduc TMT | 1.50 | 0.75 | 0.1722 | 12 | 64 | 174.2 |
| Leduc TMT | 1.50 | 0.50 | 0.0168 | 12 | 64 | 174.2 |
| Leduc TMT | 2.00 | 1.00 | 0.4703 | 12 | 64 | 174.2 |
| Leduc TMT | 2.00 | 0.75 | 0.4033 | 12 | 64 | 174.2 |
| Leduc TMT | 2.00 | 0.50 | 0.2378 | 12 | 64 | 174.2 |
| Derks timsTOF | 1.25 | 1.00 | 0.0173 | 12 | 64 | 384.0 |
| Derks timsTOF | 1.50 | 1.00 | 0.5422 | 12 | 64 | 384.0 |
| Derks timsTOF | 2.00 | 1.00 | 0.9268 | 12 | 64 | 384.0 |
| Derks Q Exactive | 1.25 | 1.00 | 0.0086 | 12 | 64 | 678.4 |
| Derks Q Exactive | 1.50 | 1.00 | 0.3724 | 12 | 64 | 678.4 |
| Derks Q Exactive | 2.00 | 1.00 | 0.7222 | 12 | 64 | 678.4 |
| Brunner label-free | 1.25 | 1.00 | 0.0714 | 12 | 64 | 896.0 |
| Brunner label-free | 1.50 | 1.00 | 0.5468 | 12 | 64 | 896.0 |
| Brunner label-free | 2.00 | 1.00 | 0.7960 | 12 | 64 | 896.0 |

Rejection summaries absent from older non-TMT grid columns are computed from their existing replicate counts. The original grid is retained unchanged; no runs were repeated.

# S5

S5, worked-example manifests.

| Series | Seed | Manifest_file | SHA256_prefix |
|---|---|---|---|
| Leduc TMT | 20261013 | leduc2022_pSCoPE-worked-example-manifest.json | 7e6fb3845a254dfb |
| Derks timsTOF | 20261013 | derks2022_timsTOF-worked-example-manifest.json | c7df0e2ea80bf6b0 |
| Brunner label-free | 20261013 | brunner2022-worked-example-manifest.json | bd4f268f5f3f8139 |

## S6

S6, the pilot protocol.

| Item | Record |
|---|---|
| Default design | 4 donors × 2 conditions × 20 cells (default, not a derived minimum) |
| Donor variance | Own cell type: log2 donor means in the reference condition |
| Carrier compression | Known-ratio two-proteome standard; k supplied to calibration |
| Background | Blank wells in every acquisition batch |

# S7

S7, the frozen mandate and addenda (the registered record).

| Record | Deposited_file | SHA256_prefix |
|---|---|---|
| Frozen mandate v1.1 | scpPower_mandate_v1.1.md | ac82ee30556ac018 |
| Frozen addendum v1.1.1 — M1 resumption, approved 13 October 2026 | scpPower_mandate_v1.1.md | ac82ee30556ac018 |
| Frozen addendum v1.1.2 — M1 closure and M2, approved 13 October 2026 | scpPower_mandate_v1.1.md | ac82ee30556ac018 |
| Frozen addendum v1.1.3 — M2 revision and conditional M3, approved 13 October 2026 | scpPower_mandate_v1.1.md | ac82ee30556ac018 |
| Frozen addendum v1.1.4 — M4 validation, approved 13 October 2026 | scpPower_mandate_v1.1.md | ac82ee30556ac018 |
| Frozen addendum v1.1.5 — M4 closure and TMT testability check, 13 October 2026 | scpPower_mandate_v1.1.md | ac82ee30556ac018 |